PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

KIBAHA PRIVATE SECONDARY SCHOOLS ASSOCOATION (KIPSSA)

FORM FOUR PRE – MOCK EXAMINATION – 2022

CHEMISTRY 2A

CODE: 032/2A

TIME: 2:30 Hours **JUNE, 2022**

INSTRUCTIONS

1. This paper consists of two (2) questions.
2. Answer all questions
3. Each question carries twenty five (25) marks
4. Cellular phones and any unauthorized materials are not allowed in the examination room.
5. Write your Examination Number on every page of your answer booklet (s).
6. You may use the following constants Atomic masses ; H = 1, C = 12, O = 16 , Na = 23, S = 32, Cl = 35.5 1Litre = $1dm^3$ = 1,000 cm^3 .

This paper consists of two Printed pages

1. You are provided with the following solution ;
   TZ: Containing 3.5g of impure Sulphuric acid in 500 cm^3 of solution
   LO: Containing 4g of Sodium hydroxide in 1,000 $cm^{3\ of}$ solution.
   MO: Methyl Orange indicator.

   Procedures:
   i. Fill the burette with solution TZ.
   ii. Pipette 20cm3 or 25cm^3 of Solution LO and put it in a conical flask.
   iii. Titrate TZ against LO using two drops of methyl orange indicator
   iv. Repeat the titration to obtain three more titre values.

   QUESTIONS:
   a) Indicate the table of result and compute the average volume of acid used for complete neutralization.
   b) What is the suitable indicator for the titration of the given solution? Give a reason(s) for your answer.
   c) Write a balanced chemical equation for the reaction between TZ and LO
   d) Why is it important to Swirl or Shake the contents of the Flask during the addition of the acid?
   e) …….. cm^3 of acid required …….. cm3 of base for complete reaction.
   f) Showing your procedures clearly, calculate the percentage of purity of TZ.

2. Sample R is a simple salt containing one cation and one anion
   Carefully, carryout all the experimental procedures described in Table 1 below. Record all your observation and appropriate inferences to identify the ions present in sample R.

TABLE 1

S/N	Experiment	Observation	Inference
(a)	Observe sample R		
(b)	Put a spatula full of sample R in a test tube and add distilled water		
(c)	Transfer 0.5g of sample R in a test tube, add dilute Hcl acid		
(d)	Transfer 0.5g of sample R in a test tube, add concentrated sulphuric acid.		
(e)	Dissolve sample R then divide the resulting solution into three (3) Portions: i. To the first portion add sodium hydroxide solution		
	ii. To the second portion, add ammonia solution.		
	iii. To the third portion , add potassium ferricyanide solution ($K_3Fe(CN)_6$)		
	iv. To the fourth , portion , add lead acetate solution followed by acetic acid solution		

Conclusion:
   a) (i) The action in sample R is ……..
      (ii) The anion in Sample R is ……
      (iii) The compound R is ……..
   b) Write the reaction equation that took place at experiment (c)
   c) State two chemical properties of each non – metal present in sample R.
   d) State one use of R.